Fiscal Year:	FY 2026	**Task Last Updated:**	FY 05/11/2026
PI Name:	Lan, Li-i		
Project Title:	Ultra-Compact Urinary Calcium Measurement Device: Refinement and Application		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	Yes
Human Research Program Elements:	(1) **ExMC**:Exploration Medical Capabilities		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	03755	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2020 HERO 80JSC020N0001-FLAGSHIP, OMNIBUS1 Human Research Program: Crew Health Appendix A; Omnibus1-Appendix B
Start Date:	03/28/2022	**End Date:**	09/27/2025
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:	NOTE: End date has been changed to 09/27/2025 per NSSC information (Ed., 7/12/24). NOTE: End date has been changed to 09/27/2024 per NSSC information (Ed., 11/15/23). NOTE: End date has been changed to 09/27/2023 per V. Lehman/JSC (Ed., 3/24/23). Original end date was 03/27/2023.		
Key Personnel Changes/Previous PI:	Note: Per the Principal Investigator (PI), Lionel Lewis has been added to the list of Co-Investigators (Ed., 3/11/25).		
COI Name (Institution):	Buckey, Jay M.D. (Dartmouth College) Devoy, Clive (Creare LLC) Phillips, Scott Ph.D. (Creare LLC) Knaus, Darin Ph.D. (Creare LLC) Lewis, Lionel (Geisel School of Medicine at Dartmouth)		
Grant/Contract No.:	80NSSC22K0847		
Performance Goal No.:			

Performance Goal Text:	
Task Description:	Slowing bone loss and preventing kidney stone formation are critical for successful spaceflight. The capability to track bone loss and kidney stone risk while in space would allow these risks to be monitored directly and would enable individualized countermeasure programs. At present, post-flight measurements are used to establish the effectiveness of the in-flight bone loss/kidney stone prevention program. A preventive approach would offer much greater operational flexibility, where ongoing in-flight measurements of countermeasure effectiveness would allow for adjustments in the countermeasure program during the flight. This approach could also be used during times when countermeasure equipment is broken, or when scheduling impacts the countermeasure program, to assess how this is affecting bone loss and kidney stone risk. In a previous NASA Omnibus project, we developed a prototype compact, low-power system to make urinary calcium measurements in space. This approach has proven to be feasible, but further development is needed to advance toward flight use. The objective of the current project is to improve this ultra-compact, robust, urinary calcium measurement system. This study will further a technology that may offer a personalized, preventive approach to bone loss and kidney stone prevention in space. --- A second project was added to this grant. This augmentation was focused on studying the pharmacokinetics of intranasal scopolamine to determine if it can provide a rapid time to maximum concentration (Tmax). Fast relief from motion sickness during spaceflight is critical to crew well-being and performance. Scopolamine is an effective drug for motion sickness and intranasal administration may offer a rapid mode of administration. Also, intranasal administration offers the ability to self-administer the drug while suited, for either prevention or treatment. A gel formulation for intranasal administration has been evaluated but has a slow onset of action, on the order of 60 minutes. Our pilot work at Dartmouth College showed that nebulized intranasal administration of an aqueous scopolamine formulation may have an onset as fast as 5 minutes. The project goal was to test the pharmacokinetics of our nebulized intranasal aqueous scopolamine formulation on 14 subjects by obtaining 12 serial blood samples at several intervals over 3 hours.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research benefits life on Earth by developing a low cost, portable, and simple way to test urine calcium concentration in a remote setting or from the comfort of one's own home. This can be used for early detection of kidney disease. At-home testing of urinary calcium biomarkers allows individuals to monitor their kidney function regularly. This can help detect kidney disease at an early stage, thus enabling timely treatment and preventing further damage. A device with this portability and form factor promotes convenience. At-home testing of urinary calcium biomarkers is quick, easy, and can be done in the privacy of one's own home. This convenience can encourage more individuals to monitor their kidney function regularly. The simplicity of this device can also be cost-effective. At-home testing of urinary calcium biomarkers is less expensive than going to a healthcare provider for the same test. This cost savings can increase access to kidney health testing for individuals who cannot afford traditional healthcare services. This in turn has public health benefits. Regular monitoring of urinary calcium biomarkers can help identify populations at risk for certain health conditions, including kidney disease, and allow for targeted public health interventions. Overall, a low cost, simple, and portable device for measuring urinary calcium biomarkers can promote early detection of kidney disease, provide convenience and cost savings, and have significant public health benefits. --- Research into fast acting motion sickness relief can provide more ways to manage motion sickness in the general population. The fast-action and non-invasive administration of intranasal nebulized scopolamine provides an attractive alternative to many oral medications.
Task Progress:	Task Progress: This reporting period marks the completion of work under this grant. The urinary calcium device project was completed in an earlier reporting period, with detailed results reported in the FY2024 entry and the final report submitted. This entry summarizes the final results of the intranasal scopolamine augmentation. TASK RESULTS: Intranasal Scopolamine Pharmacokinetics We enrolled 15 participants in the intranasal scopolamine pharmacokinetic study under an Institutional Review Board (IRB)-approved protocol. Twelve participants received a pH-adjusted aqueous scopolamine formulation via a handheld, compact nebulizer developed by Creare LLC. Two additional participants received the same formulation via an atomizing nasal spray to compare delivery methods. For both methods, half of the dose was delivered to each nostril, and subjects held their breath during administration. Following dosing, blood samples were collected at 12 time points over a 3-hour window (5, 10, 15, 20, 25, 35, 45, 60, 90, 120, 150, and 180 minutes post-dose) and analyzed in batch by mass spectrometry. At each time point, subjects also rated 10 potential side effects on a 0–10 scale and had blood pressure and heart rate measured. In the nebulizer group, one participant had no detectable scopolamine in the blood, likely reflecting a problem with drug preparation or delivery. For the remaining subjects, the nebulizer produced a median Tmax of 8.6 minutes and a mean Cmax of 938.5 pg/mL. The nasal spray produced a median Tmax of 7.5 minutes and a mean Cmax of 1179 pg/mL. By comparison, an intranasal gel formulation evaluated under an earlier project had a median Tmax of 45 minutes and a mean Cmax of 273.6 pg/mL. Because the administered dose was the same across all three devices, the higher Cmax values for the nebulizer and spray suggest more efficient absorption of the aqueous formulation. Reported side effects were consistent with the known scopolamine profile, with dry mouth, blurred vision, dizziness, and drowsiness rated highest in both groups. One of the two spray subjects reported significant disorientation/confusion; the other reported none. Blood pressure was unremarkable across the study, and heart rate showed a small but consistent drop following drug administration. CONCLUSION:

	The augmentation portion of this project successfully identified a drug and delivery combination that can provide non-invasive, rapid-onset motion sickness relief. Nebulized aqueous scopolamine produced drug onset rates comparable to intravenous administration, and preliminary data from the nasal spray subgroup were comparable to the nebulizer. Because the nasal spray device is simpler, more compact, less expensive, and easier to use than a nebulizer, it is an attractive candidate for further development. Future steps include expanding the spray cohort from n=2 to n=12 to confirm the preliminary findings, testing other motion sickness medications such as promethazine in the same formulation and delivery format, and evaluating nebulized or sprayed scopolamine in operational settings. This work directly addresses the NASA Human Research Roadmap gap related to space motion sickness countermeasures.
Bibliography Type:	Description: (Last Updated: 04/27/2026)
Articles in Peer-reviewed Journals	Buckey JC, Lan M, Lewis LD. "Space travel-associated motion sickness and its treatment." British Journal of Clinical Pharmacology. 2026 Jan;92(1):5-10. https://doi.org/10.1002/bcp.70056 ; PMID: 40183157 , Jan-2026